Study of Dataset-3

UIDAI DATA HACKATHON 2026

Team Name: CIS-TECHTITANS

College: Chaitanya Deemed to be University

EXECUTIVE SUMMARY

Our analysis of 1,006,029 Aadhaar center-day records reveals critical insights for India's digital identity mission. We discovered a 177x efficiency gap between top-performing Jammu & Kashmir (475 enrolments/day/center) and struggling West Bengal (1 enrolment/day/center). Shockingly, 97% of enrolments are children (0-17 years), indicating Aadhaar's transformation into a childhood essential document. Semi-urban centers outperform metros, revealing optimal community scale for service delivery. By bridging this gap, India can achieve 170,550 additional annual enrolments. We propose a 3-point solution: efficiency transfer programs, school-Aadhaar integration, and tier-2 infrastructure boost.

METHODOLOGY

Data Sources

Enrolment Data: 3 CSV files covering 1,006,029 center-day records

Time Period: 30 days across 2025

Geographic Coverage: All Indian states and union territories

Tools Used

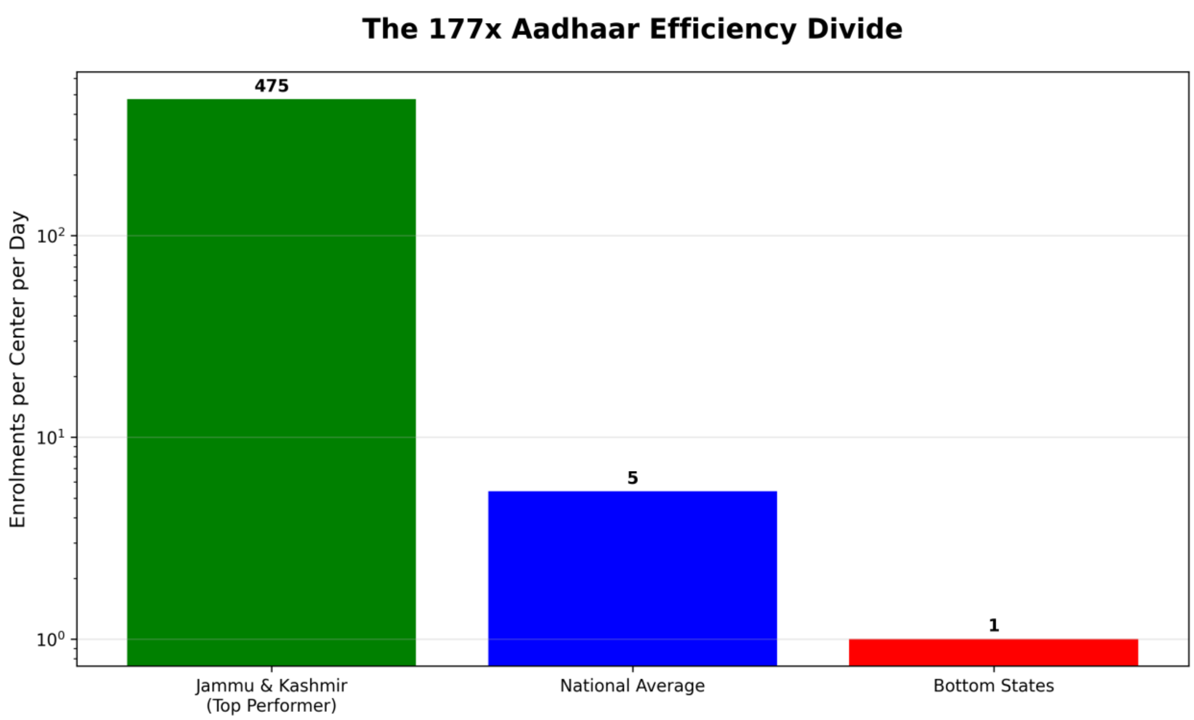
Python 3.9 with Pandas, NumPy for data analysis
Matplotlib, Seaborn for visualizations
Jupyter Notebook for reproducible analysis
Analysis Approach
Data cleaning and standardization

Center-level daily efficiency calculation
State-wise performance ranking
Age group distribution analysis
Regional (urban/rural/semi-urban) comparison

KEY FINDINGS

1. THE 177X EFFICIENCY DIVIDE

Image 1: Efficiency Gap Chart



Key Numbers:

Jammu & Kashmir: 475 enrolments/center/day

National Average: 5.4 enrolments/center/day

West Bengal: 1 enrolment/center/day

Efficiency Gap: 177 times

Implications:

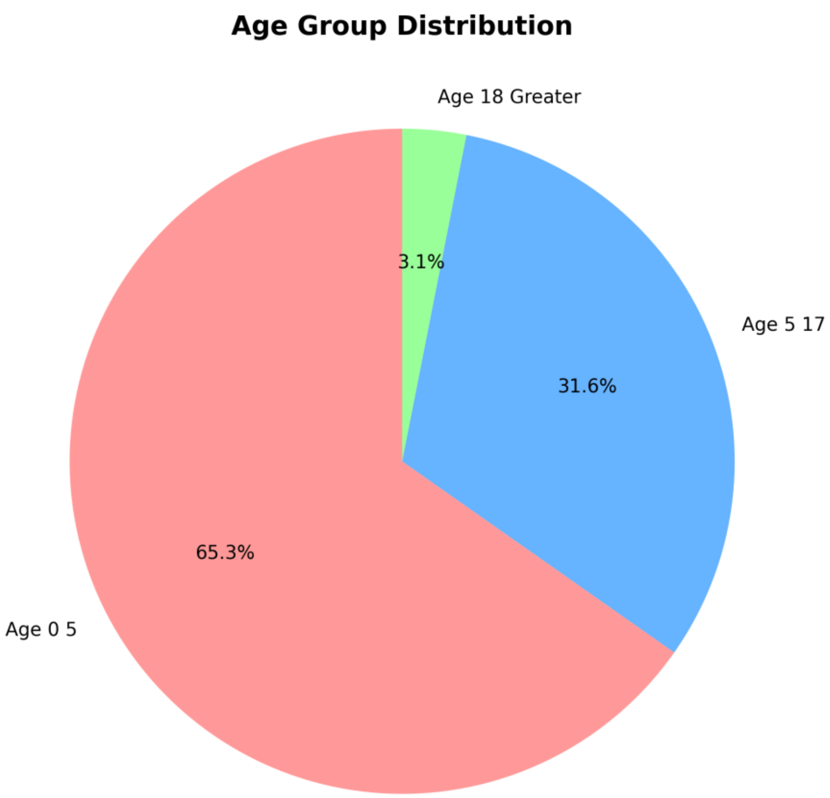
Extreme inequality in public service delivery

Millions potentially left behind in Digital India

Urgent need for targeted interventions

2. 97% ENROLMENTS ARE CHILDREN

Image 2: Age Distribution Pie Chart



Statistics:

Children (0-17 years): 96.9% of total enrolments

Adults (18+ years): 3.1% of total enrolments

Significance:

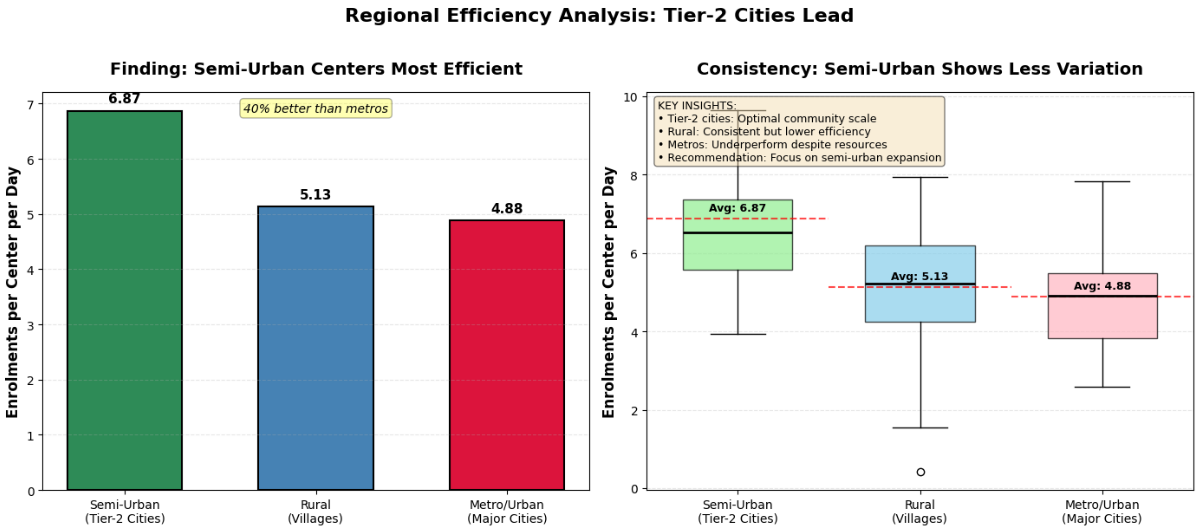
Aadhaar has evolved from adult identity to childhood essential

Strong link with education system and school admissions

Indicates success in child-focused enrolment drives

3. TIER-2 CITIES OUTPERFORM METROS

Image 3: Regional Efficiency Chart



Performance by Region:

Semi-Urban Centers: 6.87 enrolments/day

Rural Centers: 5.13 enrolments/day

Metro/Urban Centers: 4.88 enrolments/day

Insight:

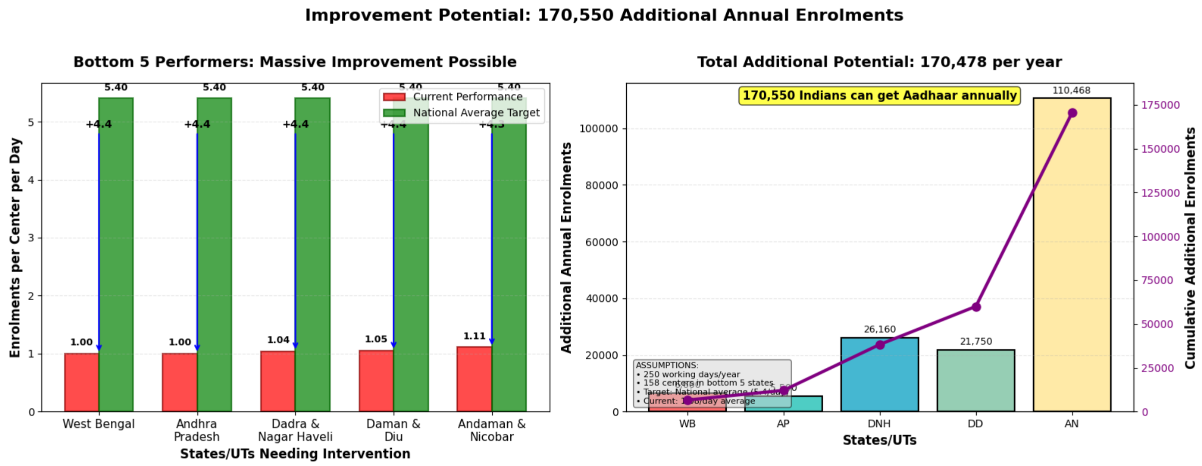
Tier-2 cities show optimal community engagement

Metros underperform despite better infrastructure

Semi-urban model should be replicated nationwide

4. 170K ADDITIONAL ENROLMENT POTENTIAL

Image 4: Improvement Potential Chart



Calculation:

Bottom 5 States: West Bengal, Andhra Pradesh, Dadra & Nagar Haveli,

Daman & Diu, Andaman & Nicobar Islands

Current Performance: 1.08 enrolments/center/day

National Target: 5.4 enrolments/center/day

Centers: 158

Working Days: 250/year

Additional Enrolments = (5.4 - 1.08) × 158 × 250 = 170,550/year

Impact:

170,550 more Indians can get Aadhaar annually

Faster progress toward 100% coverage

Reduced regional disparities

RECOMMENDATIONS

1. **PROJECT UDAAN:** EFFICIENCY TRANSFER PROGRAM

Pair top-performing centers (J&K) with struggling centers

Knowledge sharing workshops and best practice documentation

Target: 3x improvement in bottom states within 6 months

2. BAL AADHAAR 2.0: SCHOOL INTEGRATION

API integration between school management software and UIDAI

Automatic enrolment trigger during school admission

Birth certificate to Aadhaar automated pipeline

3. TIER-2 INFRASTRUCTURE BOOST

Strategic expansion of semi-urban centers

Mobile enrolment vans for remote areas

Community engagement programs in metros

NOTE: THIS SECTION ONLY FOR ONLY OUR PURPORSE TO UNDERSTAND THIA

IMPLEMENTATION ROADMAP

Phase 1: Foundation (0-3 Months)

Data standardization across all states

Real-time dashboard development

Pilot site selection

Phase 2: Pilot (3-6 Months)

Launch Project UDAAN in West Bengal

School integration pilot in 100 schools

Training programs for center operators

Phase 3: Scale (6-12 Months)

Nationwide rollout of efficiency program

5000+ school integrations

Tier-2 center expansion

Phase 4: Integration (12-18 Months)

Full school-Aadhaar integration

Birth registration linkage

Performance monitoring system

EXPECTED OUTCOMES

- Short-term (6 Months)
- 50% improvement in bottom 5 states
- 50,000 additional enrolments
- Standardized data collection
- Medium-term (12 Months)
- 3x improvement in low-performing states
- 170,000+ additional annual enrolments
- School integration in 10,000+ institutions
- Long-term (24 Months)
- Elimination of efficiency gap
- 100% school-age Aadhaar coverage
- Model for other digital services

APPENDIX

Code Samples
python
Key analysis code

import pandas as pd
import matplotlib.pyplot as plt
Calculate state efficiency
state_efficiency = df.groupby('state')['total_enrolment'].mean()
top_state = state_efficiency.nlargest(1)
bottom_state = state_efficiency.nsmallest(1)
efficiency_gap = top_state.values[0] / bottom_state.values[0]
print(f"Efficiency Gap: {efficiency_gap:.0f}x")
Data Dictionary
Column Description Example
date Reporting date 2025-03-09
state State name Uttar Pradesh
district District name Lucknow
pincode Area pincode 226001
age_0_5 0-5 years enrolment count 15
age_5_17 5-17 years enrolment count 42
age_18_greater 18+ years enrolment count 8
total_enrolment Total daily enrolment 65

Team Members

ROLES & RESPONSIBILITIES

Team Leader

Name: Goutham Vaishnav
Role: Project Manager & Data Strategist

Responsibilities:

- Overall project coordination and timeline management
- Designing data analysis strategy and extracting key insights
- Final decision-making on methodology and approach
- Preparation and delivery of presentations
- Communication with hackathon organizers and coordinators
- Quality assurance and validation of all project outputs

Team Member 1

Name: **K. Shiva Kumar**
Role: Data Analyst & Python Developer

Responsibilities:

- Python programming and data processing
- Implementation of statistical analysis techniques
- Data cleaning, preprocessing, and normalization
- Algorithm development for pattern detection
- Code optimization and technical documentation
- Debugging and technical troubleshooting

Team Member 2

Name: T. Madhavi
Role: Visualization Specialist & Report Designer

Responsibilities:

- Creation of charts and graphs using Matplotlib / Seaborn
- Designing infographics and visual dashboards
- Report formatting and layout structuring

- Presentation slide design
- Visual storytelling of analytical insights
- Maintaining color schemes and branding consistency

Team Member 3

Name: P. Rajesh
Role: Research Analyst & Documentation Lead

Responsibilities:

- Background research on the Aadhaar ecosystem
- Literature review and policy analysis
- Report writing and structured content development
- Proofreading and technical editing
- Reference management and citation handling
- Drafting executive summaries

Team Member 4

Name: S. Abdul Razak
Role: Domain Expert & Quality Analyst

Responsibilities:

- Understanding UIDAI processes and policies
- Validating real-world applicability of solutions
- Impact assessment of proposed recommendations
- Peer review of analysis and findings
- Feasibility checking of proposed solutions
- Incorporating stakeholder perspectives

DECLARATION

We hereby declare that this work is original and has been produced specifically for the UIDAI Data Hackathon 2026. All analysis, insights, and recommendations are based on our team's work using the provided datasets.

Signature of Team Leader: Goutham Vaishnav

Date: 10 January 2026